Freestanding Deck Construction Plan – 31 ft x 21 ft

Blue Springs, Missouri | Complete Buildable Design, Materials, Cut Lists, and Cost Estimates

Prepared July 15, 2026

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1. Project Summary

Item	Value
Location	Blue Springs, Missouri (Jackson County)
Deck type	Freestanding (fully self-supporting, not attached to house)
Finished outside dimensions	Exactly 31 ft 0 in x 21 ft 0 in
Finished deck surface height	Exactly 24 in above finished grade
Ground	Flat and level, no obstructions
Stairs / guardrails / roof / hot tub	None
Framing lumber	Pressure-treated Southern Yellow Pine (SYP) No. 2, ground-contact (GC / UC4A) rated
Posts	6x6 pressure-treated GC, on galvanized standoff bases over concrete piers
Beams	Dropped (beneath joists), built-up 3-ply 2x8
Joists	2x8 at 16 in on center, two lapped runs over a center beam
Decking options	Option A: 5/4x6 PT; Option B: 2x6 PT (both compared; recommendation in Section 15)
Decking direction	Parallel to the 31-ft dimension, picture-frame border, one center breaker board
Footings	15 poured concrete piers, bottom at 36 in below grade (local frost depth)

Because guardrails are not required where the walking surface is 30 in or less above grade (IRC R312.1.1), the 24-in height makes this a legal no-guardrail deck.

Read Section 17 (Required Cautions) and Section 18 (Consistency Audit) before buying anything or digging.

2. Code Research for Blue Springs, Missouri

Date all code information below was checked: July 15, 2026.

2.1 Requirements confirmed from current Blue Springs / Missouri sources

1. **Adopted code editions (confirmed).** The City of Blue Springs Codes Administration Division states that the City has adopted the **2024 International Residential Code (IRC)**, 2024 IBC, 2024 IFC, 2024 IPC, 2024 IMC, 2024 IEBC, the 2023 National Electrical Code, and ICC A117.1-2017. These updated codes **took effect June 1, 2026** after a 90-day transition period. Permit applications submitted on or after June 1, 2026 must comply with the newly adopted codes – which includes this project. The 2018 IECC/energy chapter

is adopted for reference only (not enforced). Source: City of Blue Springs Codes Administration Division, https://bluespringsgov.com/222/Codes-Administration-Division (adopting ordinance: Ord. No. 5480, passed February 18, 2026, https://ecode360.com/BL3243/laws/LF2617498.pdf).

2. **Frost line depth: 36 inches (confirmed in the codified municipal code).** Blue Springs Municipal Code Section 500.010(A)(8)(a) amends IRC Table R301.2(1) and sets: Frost Line Depth **36 in**; Ground Snow Load **20 psf**; Ultimate Wind Speed **115 mph**; Seismic Design Category **A**; Weathering **Severe**; Termite **Moderate-Heavy**. Source: https://ecode360.com/28249653 (Chapter 500, Article I). *Caution:* the online codified text still reflects the pre-June-2026 (2018-code) ordinance language; Ordinance 5480 re-adopts under the 2024 codes and its full amendment text is not yet published online. The 36-in frost depth is the KC-metro standard and is very unlikely to have changed, but **confirm it when you contact the City** (Codes Administration: 816-228-0118).

3. **Permit status for this specific deck (confirmed language, but must be verified by phone).** Blue Springs Municipal Code Section 500.020(A)(1)(b)(5) exempts from building permits: *"Platforms, decks and walks without any overhead structure that are not more than thirty (30) inches above grade and not over any basement or story below."* Source: https://ecode360.com/28249653. This 24-in freestanding deck with no roof appears to qualify for that exemption. **However**, the City's own public guidance ("Top 10 Code Tips") states that deck construction requires a city building permit (https://bluespringsgov.com/222/Codes-Administration-Division), and Ordinance 5480 (effective June 1, 2026) amended the permit article and its full text is not published online. **Call 816-228-0118 before starting and get the answer in writing.** Even if no permit is required, the deck must still be built to code and must respect zoning setbacks, easements, and stormwater rules.

4. **Inspection scheduling (confirmed).** Effective June 1, 2026, inspection requests must be made by 3 p.m. the business day before the requested date (Building Inspection Hotline 816-228-0104). If work is not ready when the inspector arrives the inspection can be recorded as failed and a reinspection fee may apply. Source: https://bluespringsgov.com/222/Codes-Administration-Division.

5. **Permit/application portal (confirmed).** Applications are submitted through the City's Permit and Application Center (https://www.bluespringsgov.com/1603/Permit-Application-Center) or the EnerGov self-service portal.

2.2 Standard IRC requirements applied in this design (not Blue-Springs-specific)

These come from the IRC (2021/2024 deck provisions, Section R507, whose 40 psf live-load span values are unchanged in substance from the 2018 tables the City previously enforced). Where the 2024 IRC is cited, verify the exact table values with the City plan reviewer, since Anthropic's assistant compiled them from the 2018/2021 published tables and the 2024 edition's deck tables.

- Deck live load 40 psf; deck dead load 10 psf assumed (IRC Table R301.5). Ground snow load 20 psf is less than 40 psf live, so the live load governs the design.
- Deck joist spans: IRC Table R507.6 (2x8 SYP No. 2 at 16 in o.c.: 11 ft-10 in max span; cantilever up to 1/4 of the actual backspan).
- Deck beam spans: IRC Table R507.5 (3-ply 2x8 SYP No. 2 supporting an 8.5-ft effective joist span: approx. 8 ft-11 in allowable span by interpolation between the 8-ft and 10-ft columns; we use 7 ft-0 in).
- Footings: IRC R507.3; minimum bearing 12 in below undisturbed grade and below frost depth (locally 36 in); sized for soil bearing capacity.
- Presumptive soil bearing: IRC Table R401.4.1 gives 1,500 psf for clay/silt soils – the conservative default used here (see Section 2.3).
- Guards not required: walking surface less than 30 in above grade (IRC R312.1.1).
- Post base/uplift: posts restrained laterally and against uplift at base and cap (IRC R507.4); freestanding decks are exempt from house-attachment lateral-anchor requirements but must be stable as a self-supporting structure.
- Wood protection: AWPA UC4A ("Ground Contact") treatment where wood is within 6 in of ground, is hard to replace, or has poor ventilation/drainage (IRC R317; AWPA U1). Because this entire frame sits 5 in to 16 in above the gravel, ground-contact treatment is specified for ALL framing, posts, and beams. Field-cut ends must be treated with copper naphthenate (AWPA M4).

- Fasteners and connectors in contact with treated wood: hot-dip galvanized (ASTM A153 / G185) or stainless steel (IRC R317.3). Do not mix aluminum with treated lumber.

2.3 Conservative assumptions that Blue Springs must confirm before construction

#	Assumption	Why conservative	Confirm with City
1	Soil bearing capacity 1,500 psf	IRC default for clay; Jackson County has high-plasticity clay	Ask if 1,500 psf is accepted or if a higher presumptive value (e.g., 2,000 psf) is allowed; footing diameters shrink if so
2	Footing diameters 18 in / 20 in	Sized for 1,500 psf	Final diameters and any required inspection of open holes
3	Frost depth 36 in	Codified value; re-adoption text not yet online	Confirm 36 in still applies under Ord. 5480
4	Permit exemption for a 24-in freestanding deck	Codified exemption vs. city tip-sheet conflict	Get written confirmation whether a permit is required
5	No prescriptive lateral bracing required at 24 in height	Knee braces are physically impossible with 5-in posts; load path provided instead (Sec. 5.10)	Confirm the anchored-pier load path is acceptable
6	2024 IRC deck-table values match the 2018/2021 values used here	40 psf live-load rows historically unchanged	Plan reviewer confirms table values

No local code requirements have been invented; every Blue Springs item above links to a city source, and everything else is labeled IRC-standard or assumption.

3. Design Loads and Engineering Basis

Parameter	Value	Basis
Live load (L)	40 psf	IRC Table R301.5, deck
Dead load (D)	10 psf	Framing + PT decking (actual approx. 7.5-9.5 psf; 10 used)
Total design load	50 psf	D + L
Ground snow load	20 psf	Blue Springs amendment; < 40 psf so live governs
Deflection limit	L/360 live	IRC R301.7
Soil bearing (assumed)	1,500 psf	IRC Table R401.4.1 clay default – **confirm locally**
Frost depth	36 in	Blue Springs Sec. 500.010(A)(8)(a)
Lumber	SYP No. 2, ground contact (UC4A), wet-service	Retail-standard PT

4. Structural Design and Calculations

Coordinate convention used throughout: the deck is a 31 ft (X axis, 372 in) by 21 ft (Y axis, 252 in) rectangle. **Origin (0,0) is the southwest corner.** Decking runs in the X direction; joists run in the Y direction; beams run in the X direction.

4.1 Framing concept

- Three dropped built-up beams run the 31-ft direction at Y = 2 ft-0 in, 10 ft-6 in, and 19 ft-0 in (centerlines).
- Joists (2x8 at 16 in o.c.) run the 21-ft direction in **two runs**: each joist line is two 10 ft-9 in pieces that lap over the center beam – exactly what you asked for instead of 22-ft lumber.
- Joists cantilever past the two outer beams to carry the rim; beams cantilever past the end posts. Both cantilevers are small and code-compliant.
- Five 6x6 posts per beam (15 total) on 15 concrete piers.

4.2 Joist design (shown, not just asserted)

Joist span between beam centerlines: 10.5 ft - 2.0 ft = **8 ft-6 in** (both bays identical by symmetry).

- IRC Table R507.6, SYP No. 2, 40/10 psf, wet service: 2x8 at 16 in o.c. allows **11 ft-10 in** » 8 ft-6 in. **OK, with a 39% span margin.**
- Cantilever: the frame is inset 1 1/2 in from the deck edge for the fascia (Sec. 7), so the joist tip is at the inside face of the rim, 3 in from the deck edge. Actual cantilever from the outer-beam centerline = 24 in - 3 in = **21 in = 1 ft-9 in**. Allowable = backspan/4 = 102 in / 4 = 25.5 in = **2 ft-1 1/2 in. OK.**
- Stiffness check (the "modest upgrade" you asked for): the span table is built to L/360 at the 11 ft-10 in limit. Deflection scales roughly with (span ratio)^3 for the same section, so at 8 ft-6 in this floor performs near **L/(360 x (142/102)^3)** ~ **L/970** – roughly 2.7x stiffer than a bare-minimum code deck. This is why 16 in o.c. is recommended over 12 in o.c. (full comparison in Section 15).
- Why not 2x6 joists? 2x6 at 16 in o.c. allows only 9 ft-0 in (6% margin), gives a springy feel, and a 21-in cantilever on a 2x6 is at the edge of good practice. Why not 2x10? Nothing to gain – the beam layout, not the joist, sets the span, and 2x10 wastes 2 in of the very tight 24-in height budget.

Joist count: joist lines at X = 2 1/4 in (end joist), then 16 in, 32 in, ... 368 in (23 interior lines), then X = 369 3/4 in (end joist) = **25 lines**, each made of two pieces = **50 joist pieces**, cut 10 ft-9 in from 12-ft stock (the 15-in offcuts become the beam-line blocking – Section 10). Each piece runs from the inside face of a rim (3 in from the deck edge) to 6 in past the center-beam centerline; the two pieces lap 12 in over the center beam (min. 3 in lap required by R502.6.1; 12 in provided, offset one joist thickness at alternating sides of the layout line so the 16-in decking-support grid is maintained on both halves).

4.3 Beam design

Tributary joist span carried by each beam:

- Center beam B: half of each 8.5-ft bay = 4.25 + 4.25 = **8.5 ft effective joist span**.
- Outer beams A and C: half of one bay + full cantilever = 4.25 + 1.75 = **6.0 ft**.

Design all three beams for the worst case (8.5 ft) so all beams are identical (fewer mistakes, bulk pricing).

- IRC Table R507.5, SYP No. 2, 3-ply 2x8: allowable beam span is 9 ft-5 in at 8-ft joist span and 8 ft-5 in at 10-ft joist span; interpolated at 8.5 ft: approx. **8 ft-11 in**. Post spacing used: **7 ft-0 in. OK, 27% margin.**
- Beam cantilever past end posts: posts at X = 1 ft-6 in and 29 ft-6 in; each beam runs X = 2 in to 30 ft-10 in (30 ft-8 in long), passing beneath – and supporting – the end joists at X = 2 1/4 in and 369 3/4 in. Actual cantilever = 18 in - 2 in = **1 ft-4 in**. Allowable = span/4 = 84/4 = **1 ft-9 in. OK.**
- Why 3-ply 2x8 rather than 2-ply 2x10 or 3-ply 2x10? (a) A 2x8 beam saves 2 in of depth, which this 24-in-tall deck badly needs (Section 4.6); (b) a 3-ply beam is 4.5 in wide and matches a standard Simpson BCS2-3/6 post cap on a 6x6 with no shims;
  (c) 2x8x16 stock is cheap and always in stock. A 2-ply 2x10 also passes on spans but fails (a) and (b).

Built-up beam fastening schedule (required): fasten plies with 10d (3 in) hot-dip galvanized ring-shank nails in **2 rows at 16 in o.c., staggered, from each face** (outer plies nailed to the middle ply from both sides), plus

3 nails clustered within 6 in of every ply end. Every ply butt joint must land over a post. Ply layout per beam (splices over posts only):

Ply	Pieces (cut length)	Splice locations (X)	Buy from
1 (outer)	15 ft-4 in + 15 ft-4 in	15 ft-6 in (center post)	2 pcs 2x8x16
2 (middle)	8 ft-4 in + 14 ft-0 in + 8 ft-4 in	8 ft-6 in and 22 ft-6 in posts	2 pcs 2x8x10 + 1 pc 2x8x16
3 (outer)	15 ft-4 in + 15 ft-4 in	15 ft-6 in (center post)	2 pcs 2x8x16

Adjacent plies never splice over the same post. Per beam: 5 pcs 2x8x16 + 2 pcs 2x8x10. Three beams: **15 pcs 2x8x16 and 6 pcs 2x8x10.**

4.4 Posts

- 15 posts: 5 per beam at X = 1 ft-6 in, 8 ft-6 in, 15 ft-6 in, 22 ft-6 in, 29 ft-6 in (7 ft-0 in centers), on beam lines Y = 2 ft-0 in, 10 ft-6 in, 19 ft-0 in.
- 6x6 PT SYP ground-contact. IRC R507.4 allows a 6x6 up to 14 ft tall for any deck this size – at roughly 5 in tall, capacity is not remotely in question (a 6x6 SYP No. 2 short column is good for well over 20,000 lb; max post load here is 2,975 lb).
- **Post height: these are short blocks, cut to fit in the field** (Section 4.6): approx. 5 in (Option A decking) or 4 1/2 in (Option B) with the specified 2 1/2-in pier reveal. Cut each one individually after the piers are poured so pier tolerance never changes the finished 24-in deck height.

4.5 Footing design (calculations shown)

Worst-case post load (center beam, interior post): tributary area = 8.5 ft (joist trib) x 7.0 ft (post spacing) = 59.5 sq ft. Load = 59.5 x 50 psf = **2,975 lb**.

Outer-beam interior post: 6.0 ft x 7.0 ft = 42.0 sq ft x 50 = **2,100 lb**. End posts carry about 5/7 of those values.

Required bearing area at 1,500 psf (assumed – confirm):

- Center beam piers: 2,975 / 1,500 = 1.98 sq ft -> dia = sqrt(4 x 1.98 / pi) = 19.1 in -> **use 20 in diameter (5 piers)**.
- Outer beam piers: 2,100 / 1,500 = 1.40 sq ft -> 16.0 in -> **use 18 in diameter (10 piers)** (kept uniform along both outer beams for simplicity).

If Blue Springs accepts 2,000 psf, the same math gives 16.5 in and 14 in; you could then use 18 in and 14-16 in and save roughly 45 bags of concrete. **Ask.**

- Configuration: straight cylindrical pier, full diameter top to bottom, bottom on undisturbed soil at **36 in below grade**, top at **2 1/2 in above grade** (total height 38 1/2 in). Slight dome/slope on top to shed water. No belling required. Cardboard form tubes are optional (only the top 8 in needs forming in cohesive clay).
- Thickness: the pier IS the footing (monolithic column, min 6 in thickness rule far exceeded). Plain concrete is code-acceptable for this; two #4 vertical rebar per pier is a cheap optional upgrade listed in the materials.
- Concrete: bagged high-strength (4,000 psi) mix, mixed on site as you specified. Volume: 20-in piers: area 2.182 sq ft x 3.21 ft = 7.00 cu ft x 5 = 35.0 cu ft. 18-in piers: 1.767 x 3.21 = 5.67 cu ft x 10 = 56.7 cu ft. Total **91.7 cu ft**. At 0.60 cu ft per 80-lb bag: 153 bags; +10% waste = **168 bags of 80-lb mix** (about 6.7 tons dry). *Practical note:* that is a very large hand-mix; a rented mixer is strongly advised, and a 3.5-cu-yd ready-mix delivery is usually cheaper than 168 bags – you specified bagged, so bagged is what is priced, with this flag.

4.6 Height-budget verification (24 in is tight – here is the exact stack)

Measured from finished grade (0.00 in), Option A (5/4 decking):

Layer	Thickness	Elevation (top)
Pier reveal above grade	2 1/2 in	2 1/2 in
ABU66Z standoff plate	1 in	3 1/2 in
6x6 post block	5 in (cut to fit)	8 1/2 in
3-ply 2x8 dropped beam	7 1/4 in	15 3/4 in
2x8 joist	7 1/4 in	23 in
5/4 decking	1 in	**24 in exactly**

Option B (2x6 decking, 1 1/2 in thick): identical except the post block is cut to 4 1/2 in and the joist top sits at 22 1/2 in.

Verification notes:

1. **It fits – with short post blocks.** A dropped-beam deck at 24 in necessarily has stub posts. On a 5-in stub, the ABU66Z base fasteners occupy roughly the bottom 2 1/2 in and the BCS2-3/6 cap fasteners the top 2 in; the two connectors' screw zones just clear each other (offset them in plan a half-inch as well). This was checked against Simpson's published connector dimensions – tight but buildable.
2. If your inspector or an engineer dislikes 5-in stubs, the strongest practical alternative is to **delete the post blocks entirely**: pour each pier 5 in taller (to elevation 7 1/2 in), cast a 5/8-in galvanized anchor bolt projecting 8 in, set the beam on a strip of self-adhering flashing membrane over the pier, drill the beam, and countersink a hot-dip washer and nut flush into the beam top. Same elevations, one fewer trade-off. It requires accurate anchor placement, which is why the stub-post detail is primary.
3. **Ground clearance:** underside of beams is 8 1/2 in above grade and about 5 1/2 in above the gravel surface; joist undersides are 15 3/4 in above grade. That is low – which is exactly why every framing member is specified ground-contact rated, the ground gets fabric + free-draining gravel, and all four sides remain open (no skirting) for cross-ventilation. Do not add skirting without adding vents.
4. Pier reveal tolerance: anywhere from 1 1/2 in to 3 in of reveal is fine – the post blocks are cut individually to hit the beam elevation of exactly 8 1/2 in (Option A) or 8 in (Option B) above grade. **The deck's 24-in height is locked in by the post-block cut, not by the concrete pour.**

4.7 Connections summary

Connection	Method
Pier to post base	Simpson ABU66Z on 5/8-in Titen HD screw anchor (4-in min. embed), 1-in standoff
Post base to 6x6	Simpson SD connector screws, all holes filled
6x6 to beam	Simpson BCS2-3/6 post cap (ZMAX), all holes filled
Joist to outer beams	Simpson H2.5AZ hurricane/uplift tie, one per joist per outer beam (50 total)
Joist laps at center beam	Face-nail lap with (4) 10d HDG ring-shank + 2x8 blocking each bay; toe-nail each joist to beam (2) 10d
Rim joist to joist ends	(3) 16d HDG ring-shank per joist end, plus decking screws through border
Rim splices	Butt between joists with 24-in 2x8 scab, (8) 10d each side
Blocking	(2) 10d toe-nails each end
Built-up beams	Schedule in Section 4.3

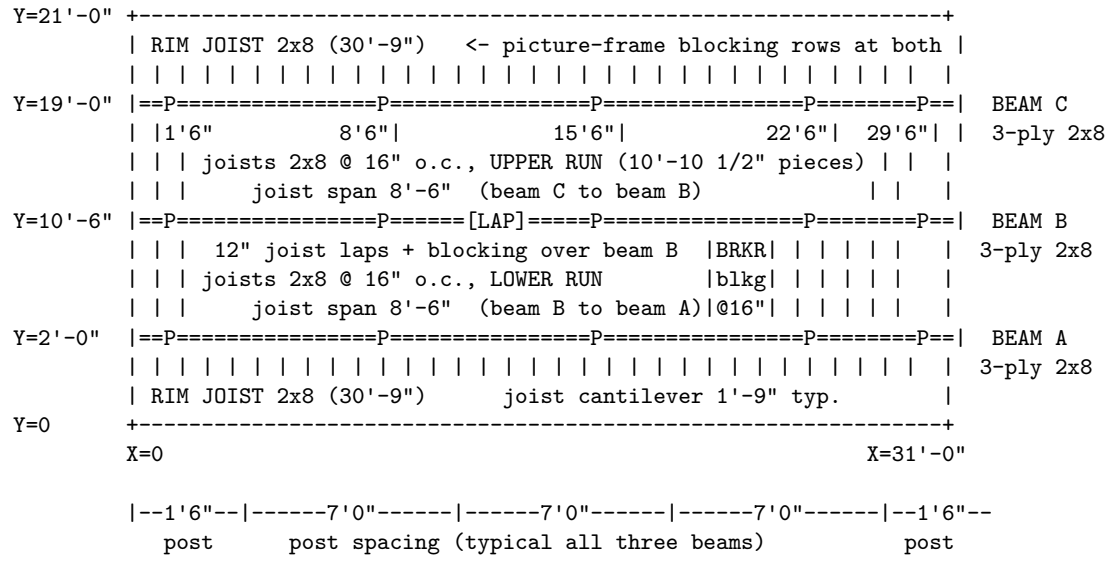
4.8 Lateral stability of a freestanding 24-in deck (bracing statement)

Diagonal knee braces are geometrically impossible on 5-in posts and are not needed at this height. Lateral and uplift stability is provided by a continuous load path: decking screwed to joists (diaphragm) -> H2.5AZ ties and blocking rows -> beams -> BCS post caps -> posts -> ABU66Z bases -> 5/8-in anchors -> 15 concrete piers totaling roughly 13,500 lb, plus about 6,500 lb of structure dead weight, all engaged 36 in below grade. Overturning

at a 2-ft height is negligible; sliding is resisted by 15 anchored piers. Full-depth 2x8 blocking is installed over both outer beams in every joist bay (48 blocks) to transfer these forces and stop joist roll. This is the standard approach for low freestanding decks; the IRC's prescriptive deck lateral-load anchors (R507.9.2) apply to house-attached decks. If the City requests engineering for the lateral system, have a licensed engineer stamp this detail (Section 17). # 5. Framing and Footing Drawings

5.1 Top-down framing plan (not to scale; all dimensions govern)

X runs left-right (31 ft-0 in outside); Y runs bottom-top (21 ft-0 in outside). Origin 0,0 = lower-left (southwest) corner. P = post/pier below beam. Deck boards run left-right (parallel to 31 ft); joists run up-down at 16 in o.c.



Key plan facts (these numbers govern over the sketch):

- Outside of FRAMING = 30 ft-9 in x 20 ft-9 in (inset 1 1/2 in on all four sides). The 1 1/2-in 2x10 fascia wraps the frame and the decking finishes flush with the fascia face, so the finished deck is exactly 31 ft-0 in x 21 ft-0 in (Sections 7-8). All coordinates below are measured from the FINISHED deck corner (0,0), not the frame.
- Beams (centerlines): Y = 2 ft-0 in (A), 10 ft-6 in (B), 19 ft-0 in (C); each beam 30 ft-8 in long, running X = 0 ft-2 in to 30 ft-10 in.
- Posts (centerlines): X = 1 ft-6 in, 8 ft-6 in, 15 ft-6 in, 22 ft-6 in, 29 ft-6 in on each beam line. Beam cantilever past end posts = 1 ft-4 in.
- Joist lines: X = 2 1/4 in, 16 in, 32 in ... 368 in, 369 3/4 in (25 lines).
- Joist cantilever beyond outer beam centerlines = 1 ft-9 in to joist tip.
- Blocking rows (2x8 full depth): over beam A (24 blocks), over beam C (24), at joist laps over beam B (24), picture-frame support between the end joist and first grid joist at each end (16 blocks per end at 16 in o.c. vertically, 12 1/4 in long), breaker-board support between joist lines X = 176 in and 192 in (16 blocks at 16 in o.c., 14 1/2 in long).
- BRKR = center breaker board zone, centered at X = 15 ft-6 in.

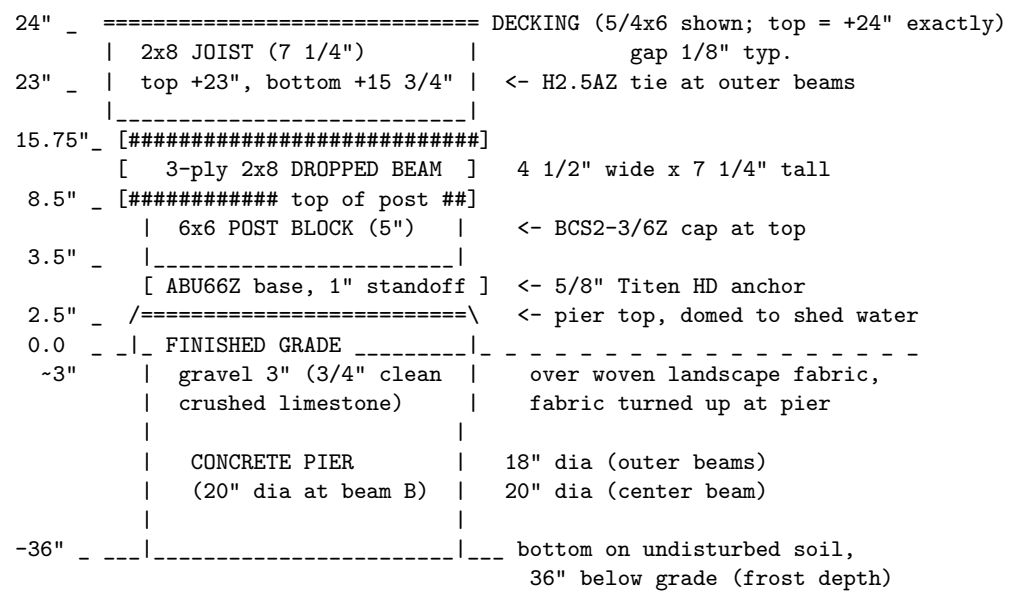
5.2 Footing coordinate table

Origin 0,0 at southwest corner; X along the 31-ft side, Y along the 21-ft side. All footings: bottom 36 in below grade, top 2 1/2 in above grade (38 1/2 in total), straight cylinder. Depth/diameter to be confirmed by Blue Springs (Sec. 2.3).

Ftg	X	Y	Beam / post	Dia	Depth below grade	C-C to neighbors
F1	1'-6"	2'-0"	A / A1	18 in	36 in	7'-0" (F2); 8'-6" (F6)
F2	8'-6"	2'-0"	A / A2	18 in	36 in	7'-0" (F1, F3); 8'-6" (F7)
F3	15'-6"	2'-0"	A / A3	18 in	36 in	7'-0" (F2, F4); 8'-6" (F8)
F4	22'-6"	2'-0"	A / A4	18 in	36 in	7'-0" (F3, F5); 8'-6" (F9)

Ftg	X	Y	Beam / post	Dia	Depth below grade	C-C to neighbors
F5	29'-6"	2'-0"	A / A5	18 in	36 in	7'-0" (F4); 8'-6" (F10)
F6	1'-6"	10'-6"	B / B1	20 in	36 in	7'-0" (F7); 8'-6" (F1, F11)
F7	8'-6"	10'-6"	B / B2	20 in	36 in	7'-0" (F6, F8); 8'-6" (F2, F12)
F8	15'-6"	10'-6"	B / B3	20 in	36 in	7'-0" (F7, F9); 8'-6" (F3, F13)
F9	22'-6"	10'-6"	B / B4	20 in	36 in	7'-0" (F8, F10); 8'-6" (F4, F14)
F10	29'-6"	10'-6"	B / B5	20 in	36 in	7'-0" (F9); 8'-6" (F5, F15)
F11	1'-6"	19'-0"	C / C1	18 in	36 in	7'-0" (F12); 8'-6" (F6)
F12	8'-6"	19'-0"	C / C2	18 in	36 in	7'-0" (F11, F13); 8'-6" (F7)
F13	15'-6"	19'-0"	C / C3	18 in	36 in	7'-0" (F12, F14); 8'-6" (F8)
F14	22'-6"	19'-0"	C / C4	18 in	36 in	7'-0" (F13, F15); 8'-6" (F9)
F15	29'-6"	19'-0"	C / C5	18 in	36 in	7'-0" (F14); 8'-6" (F10)

5.3 Side section (looking along X, through a post line)



Vertical dimensions (Option A): grade 0; pier top +2 1/2 in; standoff +3 1/2 in; post top / beam bottom +8 1/2 in; beam top / joist bottom +15 3/4 in; joist top +23 in; **deck surface +24 in**. Fascia (2x10) face covers +13 3/4 in to +23 in. Option B: post cut 4 1/2 in; beam bottom +8 in; joist top +22 1/2 in; surface +24 in.

6. Ground Preparation

Area under deck: 31 ft x 21 ft = **651 sq ft**.

1. **Call 811 (Missouri One Call) before any digging.** Wait for all utility marks.
2. Strip grass/organics 1 in to 2 in across the footprint (deck area only; keep the surrounding grade unchanged so the 24-in reference doesn't move). Do this AFTER footings are poured and framing is up, or before layout if you prefer working on clean ground – either works; the step list in Section 12 does it after framing so spoil from the piers can be hauled out first.
3. Grade flat with a very slight fall (about 1 in over the 21 ft) away from the house side so water sheds out from under the deck; compact any disturbed soil.
4. **Landscape fabric:** heavy woven (not cheap spun) geotextile across the entire 651 sq ft, seams overlapped 12 in, turned up 2 in at each pier, pinned with 6-in staples about every 3 ft and at every seam edge. Using 4-ft x 100-ft rolls at 3-ft effective width: 7 runs x 31 ft = 217 lineal ft -> **3 rolls** (includes overlap and turn-ups; approx. 20% spare).
5. **Gravel:** 3/4-in clean (washed) crushed limestone – locally called "3/4 clean rock." Clean rock drains and discourages weeds; do NOT use minus/fines material (holds moisture) under a low deck. Depth: **3 in settled**.

Quantity: 651 sq ft x 0.25 ft = 162.8 cu ft = 6.03 cu yd; add a 10% allowance for settlement/compaction into the fabric = **6.6 cu yd, or about 9.3 tons** at 1.4 tons/cu yd. Buy bulk from a Blue Springs-area rock yard (bagged equivalent is about 360 half-cu-ft bags – impractical and roughly 3x the cost; both are priced in Section 11 for completeness).

6. **Drainage:** the site is flat; clean rock over fabric with the 1-in fall is sufficient. Do not create a bathtub – feather the gravel flush to grade at the deck perimeter, never bermed above the surrounding lawn.
7. **Edging/containment:** optional. The deck skirt area is open; if you want a crisp gravel edge at the drip line, add steel or composite landscape edging around the 104-ft perimeter (listed as an optional material). Not structurally required.
8. Spread gravel AFTER piers/posts/beams are set (protects fabric during construction and lets you fine-grade around the piers), and BEFORE decking closes off access. # 7. Decking Layout (both options)

Critical dimension control: the FRAME is built 30 ft-9 in x 20 ft-9 in outside of rims – 1 1/2 in smaller than the deck on all four sides. The 1 1/2-in-thick 2x10 fascia wraps the frame, and the decking (field, border, breaker) is installed flush with the fascia face. Result: **finished outside-to-outside decking = exactly 31 ft-0 in x 21 ft-0 in**, with every rim covered. Do not frame to 31 x 21 or the fascia will push the finished size to 31'-3" x 21'-3".

7.1 Board layout math (do this before framing – it is already done here)

Across the 21-ft (Y) direction, using 5 1/2-in actual-width boards:

- 2 border courses + 43 field courses = 45 courses x 5.5 in = 247.5 in of wood.
- 252 in - 247.5 in = 4.5 in of gap distributed over 44 gaps = **3/32 in per gap**.
- Layout works out to full-width boards everywhere – **no ripped, awkward final course**. That is why the gap is specified at 3/32 in rather than 1/8 in: at 1/8 in you would come up 1 3/8 in short at the last course and need a rip. Retail PT boards are sold wet and shrink; a 3/32-in installed gap typically opens to about 1/4 in after a season, which is ideal for drainage.
- Practical method: snap course lines every 5 19/32 in (5.5 + 3/32); check the running total every 5 courses against a tape from the border (should advance 27 31/32 in per 5 courses) and correct 1/32 in at a time. Board-width variation absorbs the rest.

Along the 31-ft (X) direction:

- Border (5.5 in) + 3/32 gap at each end; breaker board (5.5 in) centered at X = 15 ft-6 in with a 3/32 gap each side.
- Each field board run = (372 - 2 x 5.5 - 5.5 - 4 x 3/32) / 2 = **177 9/16 in (14 ft-9 9/16 in)** – cut from 16-ft boards, one clean crosscut each, no butt joints in the field. All field joints occur only at the breaker and border, exactly as specified. Offcuts (about 14 1/2 in) become spacer gauges and stakes.

7.2 Option A – 5/4x6 pressure-treated decking

- 5/4x6 (1 in x 5.5 in actual) SYP, ground-contact preferred (above-ground acceptable; GC is specified here because clearance under boards is low).
- Max joist spacing for 5/4 SYP perpendicular decking: 16 in o.c. – our layout. Do not use 5/4 on 24-in spacing.
- Fasteners: #9 x 2 1/2 in coated exterior deck screws (ACQ/CA-rated, e.g., Deck Plus, DeckMate, GRK R4 exterior, or 305 stainless). **2 screws at every joist/blocking crossing**, 3/4 in to 1 in from board edges.
- Predrill: every board end (both screws), every miter, all breaker/border screws, and any screw closer than 2 in to an end. 1/8-in pilot.

7.3 Option B – 2x6 pressure-treated decking

- 2x6 (1 1/2 in x 5.5 in actual) SYP GC. Identical course count and layout (same 5.5-in face). Structural framing is unchanged (the extra ~0.7 psf is inside the 10 psf dead load allowance).
- Fasteners: #9 or #10 x 3 in coated exterior deck screws, same pattern and predrilling.
- 2x6 could legally span 24-in joist spacing, but the frame stays at 16 in o.c. – the 16-in grid is what makes the deck feel solid (see comparison, Section 15).

7.4 Picture frame, breaker, and their support framing

- **Border on the two 31-ft sides** runs parallel to the field boards and lies across the joist cantilevers – supported by every joist at 16 in o.c.; screws into each.
- **Border on the two 21-ft ends** runs parallel to the joists, so it needs blocking: a row of full-depth 2x8 blocking at 16 in o.c. (vertically, i.e., along Y) between the end joist and the first interior joist at each end (16 blocks per end). The border and the field-board ends BOTH land on this blocking.
- **Center breaker board** runs parallel to the joists at X = 15'-6"; a row of 2x8 blocking at 16 in o.c. between the joist lines at X = 176 in and X = 192 in supports the breaker and both abutting field-board ends. Field ends bear a minimum of 1 1/2 in on the blocking.
- Border corners: 45-degree miters, predrilled, 2 screws each side of the miter. (Miters on PT open with shrinkage; acceptable, or butt the corners with the long sides running through – looks fine and moves less. Miters are shown in the cut list; the cut list works for either choice.)
- Screw the border/breaker at every bearing point (16 in o.c.) with the same screws as the field.

8. Fascia and Finish Details

- Material: **pressure-treated 2x10 SYP** (1 1/2 in x 9 1/4 in actual), ground-contact, selected straight. 2x is specified rather than 1x because it stays flat over the long 31-ft runs and matches the 1 1/2-in frame inset that preserves the exact 31 x 21 footprint.
- Coverage: top of fascia tight to the underside of the decking (elevation +23 in, Option A), face covering +13 3/4 in to +23 in – the full rim/end joist plus 2 1/2 in of shadow. The decking edge sits flush with the fascia face.
- Lengths: two 31-ft sides (16 ft + 15 ft-1 1/2 in per side, joint centered on a joist bay with a 2x8 scab already at the rim splice) and two 21-ft ends (16 ft + 5 ft-3 in per end). Corners: outside miters or butt laps (long sides run past the ends). Note the fascia corners form the true 31 ft / 21 ft outside dimension – cut to the layout, not to the lumber.
- Fastening: #8 x 2 1/2 in stainless-steel trim-head screws (stainless resists the streaking/corrosion that coated screws can show on face work), 2 screws per 16 in, 1 in from top and bottom edges, predrilled. Leave **1/8-in gaps at butt joints** and 1/16 in at miters for expansion; keep the bottom edge 5-6 in clear of the gravel.
- **End sealing (required practice, AWPA M4):** brush every field-cut or drilled surface of any PT member – fascia cuts, joist trims, post-block cuts, beam ends, countersinks – with copper naphthenate end-cut preservative (2% Cu, e.g., Copper-Green Brown or clear) before installation. Two coats on end grain.
- No stain or paint is included (per your instructions); if you ever coat it, wait until the PT reaches moisture equilibrium (typically 2-6 months).

9. Hardware Schedule

All connectors ZMAX (G185 hot-dip galvanized) or better; all structural fasteners hot-dip galvanized or Simpson SD/SDWS exterior-rated; fascia screws 305 stainless. Never combine stainless and galvanized in the same connection, and no aluminum anywhere against PT lumber. Example model numbers are Simpson Strong-Tie; USP/MiTek equivalents are acceptable if load-rated the same.

Item	Model (example)	Qty	Use / notes
Standoff post base, 6x6	Simpson ABU66Z	15	1-in standoff keeps post off concrete
Anchor for post base	Simpson Titen HD 5/8 x 6 in (THD62600H)	15 (+2 spare)	Drill-in after cure; 4-in min embed; centered in pier
Post cap, 3-ply 2x on 6x6	Simpson BCS2-3/6Z	15	Beam-to-post; fill all holes
Joist-to-beam uplift tie	Simpson H2.5AZ	50 (buy 55)	Every joist at both outer beams

Item	Model (example)	Qty	Use / notes
Connector screws	Simpson SD9112R500 (#9 x 1 1/2, 500 ct)	2 boxes	For ABU, BCS, H2.5 (use SD screws, not deck screws, in connectors)
Built-up beam nails	10d x 3 in HDG ring-shank, 10-lb box	2 (approx. 940 needed)	Beam plies (Sec. 4.3), joist laps, blocking toe-nails, scabs
Rim/frame nails	16d x 3 1/2 in HDG ring-shank, 5-lb box	1 (about 220)	Rim-to-joist ends (3 per end = 150), scabs
Deck screws, Option A	#9 x 2 1/2 in coated exterior, 25-lb pail + 5-lb box	approx. 2,700	2 per bearing; counts in Sec. 10
Deck screws, Option B	#9/#10 x 3 in coated exterior, 25-lb pail + two 5-lb boxes	approx. 2,700	Same pattern, longer screw
Fascia screws	#8 x 2 1/2 in 305 SS trim-head, 5-lb box	approx. 220	2 per 16 in
End-cut preservative	Copper naphthenate 2% (Copper-Green), quart	2	AWPA M4 field-cut treatment
Protective membrane (optional, recommended)	G-Tape/Trex Protect 4-in joist tape (65-ft rolls)	3 rolls	Cap the three beam tops (93 lf) – the wettest, hardest-to-replace members
Pier rebar (optional)	#4 x 10 ft, cut in thirds	10 sticks	2 verticals per pier
Uplift/lateral	– covered by ABU66Z + Titen HD + BCS + H2.5AZ chain (Sec. 4.8)	–	Freestanding load path
Bracing hardware	None required (Sec. 4.8)	–	Blocking rows instead

Notes: (1) BCS2-3/6Z is a common special-order item at big boxes; Menards/contractor yards often stock it – order early. (2) H2.5AZ ties install on the beam side of each joist, alternating faces so nails never share a plane. (3) Buy one Titen HD-compatible 5/8-in masonry bit. # 10. Material Schedules

All lumber: pressure-treated SYP No. 2 or better, ground-contact (UC4A). Quantities show the calculation, the minimum, and the purchase quantity including a 10% waste and overage allowance where reasonable (connectors and anchors are bought at need + noted spares instead – they don't get "wasted" like lumber). Nothing is double-counted: structure is listed once (identical for both options); each decking option is listed separately.

10.1 Structural framing (identical for Option A and Option B)

Material	Calculation	Min	Buy (+10%)
2x8x12 GC (joists)	25 lines x 2 pieces at 10'-9"	50	–
2x8x12 GC (blocking)	72 purchased blocks (32 picture-frame at 12 1/4 in, 16 breaker at 14 1/2 in, 24 lap at 13 in) at approx. 10 blocks/board; 48 beam-row blocks come free from the 50 joist offcuts (15 in each)	8	–
2x8x12 GC total	50 + 8 = 58	58	**64**

Material	Calculation	Min	Buy (+10%)
2x8x16 GC	beam plies 5/beam x 3 beams = 15; rim joists 2 pcs/side x 2 = 4	19	**21**
2x8x10 GC	beam middle-ply end pieces, 2/beam x 3	6	**7**
6x6x8 GC	15 post blocks at approx. 5 in (one 8-ft stick yields 15-16)	1	**2** (insurance + clean cuts)
2x10x16 GC (fascia)	sides: 2 pcs x 2; ends: 1 pc x 2	6	**7**
2x10x8 GC (fascia)	ends: 1 pc x 2 (yields the 5'-3" pieces)	2	**2** (spare covered by 16-ft surplus)

10.2 Concrete and footing materials

Material	Calculation	Min	Buy
80-lb high-strength (4,000 psi) concrete mix	91.7 cu ft total (Sec. 4.5) / 0.60 cu ft per bag = 153	153	**168** (+10%)
Titen HD 5/8 x 6 in anchors	15 piers	15	**17**
#4 rebar x 10 ft (OPTIONAL)	2 verticals x 15 piers x 36 in = 30 pcs x 3 ft	10 sticks	10 (optional)
Form tubes (OPTIONAL)	only if soil won't hold a clean hole; 18/20-in x 4-ft tubes	0	as needed

10.3 Ground preparation

Material	Calculation	Min	Buy
Woven landscape fabric 4 ft x 100 ft	7 runs x 31 ft = 217 lf incl. 12-in overlaps	3 rolls	**3 rolls**
Fabric staples, 6-in	approx. 1 per 9 sq ft + seams = approx. 90	90	**1 box (100)**
3/4-in clean crushed limestone	651 sq ft x 3 in = 6.03 cu yd + 10%	6.03 cu yd	**6.6 cu yd (approx. 9.3 tons), bulk**
Landscape edging (OPTIONAL)	104 lf perimeter	–	optional

10.4 Hardware and fasteners (both options except deck screws)

As scheduled in Section 9: ABU66Z x15; BCS2-3/6Z x15; H2.5AZ x55; SD9112R500 x2 boxes; 10d x 3 in HDG ring-shank x2 10-lb boxes (approx. 940 nails needed: approx. 330 beam plies + 100 laps + 480 blocking + 32 scabs); 16d x 3 1/2 in HDG ring-shank x1 5-lb box (approx. 150 rim nails); SS trim screws x1 5-lb box (approx. 220 needed); copper naphthenate x2 qt; Titen HD x17; optional beam tape x3 rolls.

10.5 Option A decking (5/4x6)

Screw count (2 per bearing): field 43 courses x 26 bearings x 2 = 2,236; border long sides 100; border ends 64; breaker 32 = **2,432 min; buy approx. 2,700** (one 25-lb pail plus one 5-lb box of #9 x 2 1/2 in; verify count on the pail label – pails run 2,000-2,600 screws).

Material	Calculation	Min	Buy (+10%)
5/4x6x16 GC (field)	43 courses x 2 boards	86	95
5/4x6x16 GC (border long sides)	2 pcs/side x 2	4	4
5/4x6x16 GC (border ends, long pc)	1/end x 2	2	2
5/4x6x16 GC (breaker, long pc)	1	1	1 + 1 spare
5/4x6x16 GC total		93	**103**
5/4x6x8 GC	border-end short pcs (2) + breaker short pc (1)	3	**4**
Deck screws #9 x 2 1/2 in	2,432 + 10%	2,432	**approx. 2,700**

10.6 Option B decking (2x6) – changes from Option A

Identical piece counts (same 5.5-in face width and layout): **2x6x16 GC x103 and 2x6x8 GC x4** replace the 5/4 boards. Structural and fastener changes caused by the heavier decking:

- No framing changes (0.7 psf extra is within the 10-psf dead-load allowance; post, beam, joist, and footing checks all still pass with the same margins).
- Deck screws become **#9 or #10 x 3 in** (same approx. 2,700 count: one 25-lb pail + two 5-lb boxes, because 3-in screws run fewer per pound).
- Post blocks are cut 1/2 in shorter (4 1/2 in) to hold the 24-in surface height.

11. Cut Lists

Kerf allowance 1/8 in is included in the "yields" logic; always cut longest pieces first. Treat every cut end with copper naphthenate.

11.1 Structure cut list (applies to BOTH options)

Material	Buy length	Cut length	Qty	Use	Offcut	Offcut reuse
2x8 GC	12 ft	10'-9"	50	joists (25 lines x 2)	15 in	YES: 48 beam-row blocks (trim to 14 1/2 in / 12 1/4 in end bays)
2x8 GC	12 ft	12 1/4 in x 32	3.2 boards	picture-frame blocking	approx. 2 in/blk	–
2x8 GC	12 ft	14 1/2 in x 16	2 boards	breaker blocking	small	–
2x8 GC	12 ft	13 in x 24	2.4 boards	lap blocking over beam B	small	shims/stakes
2x8 GC	16 ft	15'-4"	6 (2/beam)	beam plies 1 and 3, piece 1	8 in	–
2x8 GC	16 ft	15'-4"	6 (2/beam)	beam plies 1 and 3, piece 2	8 in	–
2x8 GC	16 ft	14'-0"	3 (1/beam)	beam ply 2, middle piece	24 in	24-in rim-splice scabs (4 needed)
2x8 GC	10 ft	8'-4"	6 (2/beam)	beam ply 2, end pieces	20 in	scabs/blocking spares
2x8 GC	16 ft	16'-0"	2	rim joists, piece 1 (per side)	none	–

Material	Buy length	Cut length	Qty	Use	Offcut	Offcut reuse
2x8 GC	16 ft	14'-9"	2	rim joists, piece 2 (per side)	15 in	blocking
6x6 GC	8 ft	5 in (A) / 4 1/2 in (B)	15	post blocks (cut AFTER piers cure, each to measure)	remainder	spares
2x10 GC	16 ft	16'-0"	2	fascia long sides, piece 1	none	–
2x10 GC	16 ft	15'-1 1/2"	2	fascia long sides, piece 2	10 1/2 in	–
2x10 GC	16 ft	16'-0"	2	fascia ends, piece 1	none	–
2x10 GC	8 ft	5'-3"	2	fascia ends, piece 2	33 in	spare fascia patch stock
2x10 GC	16 ft	–	1	spare (mis-cut insurance)	–	–

(Fascia pieces are cut to close the exact 31-ft / 21-ft outside dimension at the corners – cut 1/8 in long and trim to the string line.)

11.2 Option A decking cut list (5/4x6)

Material	Buy length	Cut length	Qty	Use	Offcut	Offcut reuse
5/4x6 GC	16 ft	14'-9 9/16"	86	field boards (43 courses x 2 runs)	14 7/16 in	spacer gauges, stakes; not decking
5/4x6 GC	16 ft	16'-0" (miter ends)	2	border long sides, piece 1	none	–
5/4x6 GC	16 ft	15'-0" (one miter)	2	border long sides, piece 2	12 in	–
5/4x6 GC	16 ft	16'-0" (miter one end)	2	border 21-ft ends, piece 1	none	–
5/4x6 GC	8 ft	5'-0" (one miter)	2	border 21-ft ends, piece 2	36 in	breaker backup
5/4x6 GC	16 ft	16'-0"	1	breaker, piece 1	none	–
5/4x6 GC	8 ft	4'-0 13/16"	1	breaker, piece 2 (butt over blocking)	47 in	spare border stock
5/4x6 GC	16 ft	–	9	waste/overage allowance (10% of field)	–	replaces culls and mis-cuts
5/4x6 GC	8 ft	–	1	overage	–	–

Totals check: 16-ft pcs = 86+2+2+2+1+9+1(spare in Sec. 10.5 tally) = 103; 8-ft = 4. Matches Section 10.5.

11.3 Option B decking cut list (2x6)

Identical geometry – same table as 11.2 with every "5/4x6" read as "2x6" (2x6x16 GC x103, 2x6x8 GC x4), and all screws 3 in. No other changes.

11.4 Avoiding a narrow final board (layout proof)

Course math from Section 7.1: 45 courses x 5.5 in + 44 gaps x 3/32 in = 247.5 + 4.125 = 251.625 in; the remaining 3/8 in is absorbed by setting the FIRST and LAST gap (against the picture frame) at approx. 9/32 in instead of 3/32 in – visually indistinguishable shadow lines at the border, and every course stays a full 5.5-in board. If your actual boards measure 5 1/2 in scant (common), re-run the math with the measured average width before you start screwing down: field-course count stays 43; only the gap changes. Set the border first, dry-lay 5 field courses, measure, and commit. # 12. Step-by-Step Construction Instructions

Throughout: keep a story pole marked with the key elevations (+2 1/2, +3 1/2, +8 1/2, +15 3/4, +23, +24 in for Option A) and a stringline reference. The finished elevation is protected by one rule: **post blocks are cut to measure after the concrete is in** (Step 8), so nothing before that step can ruin the 24-in height.

Step 1 – Permits, utilities, setbacks. Call Blue Springs Codes Administration (816-228-0118): confirm permit-or-exempt status (Sec. 2.1 item 3), frost depth, soil bearing value, footing sizes, and which inspections apply (typically: footing/open-hole before pour, framing, final). Check your plat for setbacks and easements – a freestanding deck usually may not sit in a utility or drainage easement. Call 811 and wait for utility locates. *Mistake to avoid:* digging before locates or pouring before an open-hole inspection (if required). *Inspection checkpoint:* written permit answer filed. *Safety:* none yet.

Step 2 – Establish the footprint. Drive stakes well outside each corner and set up batter boards. Choose the fixed corner (0,0); run a mason's line for the 31-ft side. Locate the other three corners at exactly 31'-0" and 21'-0". Mark the FRAME lines too (1 1/2 in inside the finished lines on all sides). *Measurement:* corners at (0,0), (31',0), (31',21'), (0,21'). *Mistake:* laying out the frame at 31 x 21 – the fascia then breaks the exact dimension. *Checkpoint:* self-check. *Safety:* trip hazards on lines.

Step 3 – Square the rectangle (diagonal calculation). Diagonal = sqrt(31^2 + 21^2) = sqrt(961 + 441) = sqrt(1402) = 37.443 ft = **37 ft-5 5/16 in**. Adjust corners until BOTH diagonals read 37'-5 5/16" (within 1/8 in) while the sides stay 31'-0" and 21'-0". Also check a 3-4-5 (or 9-12-15) at the origin corner. *Mistake:* measuring diagonals to stakes instead of the actual string intersections. *Checkpoint:* record both diagonals. *Safety:* –

Step 4 – Mark footing coordinates. From the batter-board lines, tape-measure each footing center from Section 5.2 (X = 1'-6“, 8'-6”, 15'-6“, 22'-6”, 29'-6“; Y = 2'-0”, 10'-6“, 19'-0”). Drop a plumb bob and drive a spike + spray-paint an 18-in or 20-in circle at each. Double-check the center beam row (Y = 10'-6") – everything laps there. *Mistake:* measuring successive 7-ft hops (error accumulates); always pull from the origin line. *Checkpoint:* 15 marked circles match the table within 1/2 in.

Step 5 – Excavate below frost depth. Dig each hole to 36 in below grade, full diameter, sides reasonably plumb, bottom undisturbed and flat (hand-clean the last 2 in; do not shovel loose soil back in). A rented one-man auger with an 8-in bit to break the center, then hand-widen, is realistic in Jackson County clay. If any hole hits fill, organic soil, or water, stop and call the City/engineer. *Mistake:* over-digging and “fixing” it with loose backfill – if you over-dig, fill with concrete, not soil. *Checkpoint:* City open-hole/footing inspection if required – schedule by 3 p.m. the prior day (816-228-0104). *Safety:* utility locates current; keep hole edges clear.

Step 6 – Form and pour the piers. Form the top 8 in of each pier (scrap tube or a site-built collar) so the top finishes at +2 1/2 in above grade, level across all 15 (check with a line level or laser – up to 1 in of variation is absorbed later by the post blocks, but flatter is better). Mix 80-lb bags to a stiff slump; rod each lift to consolidate; dome the exposed top slightly. If using the optional rebar, set two #4 verticals mid-pour. *Mistake:* soupy mix (weak, segregated); pouring on a disturbed bottom. *Checkpoint:* pier tops within 1 in of each other; photos of depth before pour. *Safety:* cement is caustic – gloves, eye protection; lift bags properly (13,400 lb total will move through your hands – pace it, or rent a mixer).

Step 7 – Set anchors and post bases. After 24-48 hr cure, re-string the beam centerlines and mark each pier center. Drill 5/8-in holes (hammer drill) and install the Titen HD anchors through the ABU66Z bases; snug, then align each base to the beam line before final torque. *Mistake:* anchor off-line – the ABU's slots give about 1/2 in of forgiveness, no more; that is why the string, not the pier, sets the mark. *Checkpoint:* all 15 bases in line and on the 7-ft grid within 1/4 in.

Step 8 – Cut and set the 6x6 post blocks (elevation control). Set a laser or dry string at the beam-bottom elevation: **+8 1/2 in (Option A) or +8 in (Option B)**. Measure from the ABU66Z seat (top of standoff plate) to the line at each pier; cut each block to ITS measurement (approx. 5 in / 4 1/2 in). Treat cuts, set blocks, drive

SD screws through every base hole. *Mistake:* cutting all 15 to the same length off one measurement. *Checkpoint:* string across all post tops reads dead level at the target elevation. *Safety:* cut 6x6 with two passes of a circular saw + handsaw; never freehand a short block against the fence – clamp it.

Step 9 – Assemble and install the dropped beams. Build each 3-ply beam ON the posts or on sawhorses beside them (it weighs roughly 200 lb): crown all plies UP, splice layout per Section 4.3 (splices over posts, adjacent plies on different posts), nail per the schedule (2 rows 10d at 16 in o.c. each face, staggered, + 3 nails at each ply end). Set each beam in its BCS2-3/6Z caps, ends at X = 2 in and 30'-10", and fill every cap hole with SD screws. Optional: cap beam tops with joist tape now. *Mistake:* splices between posts (not allowed); crowns mixed up and down. *Checkpoint:* beam centerlines at Y = 2'-0", 10'-6", 19'-0" (from the FINISHED line); tops level with each other; caps fully fastened. *Safety:* two people minimum to fly beam sections.

Step 10 – Joists, rims, blocking, bracing rows. Lay out 16-in centers on beams A and C from the origin end (first interior line at X = 16 in from the finished corner); roll out the lower-run joists (tips 3 in shy of the finished edge line = flush with the future inside face of rim), then the upper run, lapping 12 in over beam B, offset alternating sides of the line. Nail laps (4x 10d each), toe-nail at beams, install H2.5AZ at both outer beams on every joist. Cap the joist ends with the rim joists (30'-9", crowned up, 3x 16d per joist end, splice scabbed); install end joists; then blocking rows: over beams A and C every bay, lap blocking over beam B, picture-frame blocking rows at both ends, breaker blocking between X = 176 and 192 in. *Mistake:* laying out 16-in centers from the frame edge instead of the finished (0,0) corner – the decking grid and breaker blocking are dimensioned from the finished corner. *Checkpoint (framing inspection if permitted):* spacing, laps, ties, blocking all per plan – schedule before decking. *Safety:* open joist field – walk the beams, not the bays.

Step 11 – Verify dimension, level, square, elevation. Frame outside-to-outside must read 30'-9" x 20'-9"; diagonals of the frame = sqrt(30.75^2 + 20.75^2) = 37.10 ft = **37'-1 3/16"** both ways; joist tops at +23 in (A) / +22 1/2 in (B) at all four corners and mid-span. Fix anything now – decking hides everything. *Mistake:* checking level only at corners; sight the rims for humps too.

Step 12 – Ground preparation. Strip vegetation 1-2 in, fine-grade with the slight fall, remove all wood scraps and stakes from under the deck (termite bait otherwise).

Step 13 – Fabric and gravel. Roll fabric per Section 6 (12-in overlaps, staples, turn up at piers), then spread the 3 in of clean rock, raking flush at the perimeter. Working under 23-in joists is real crawling – many builders do Steps 12-13 right after Step 7 instead, before the frame goes up; either order is fine as long as the fabric isn't torn during framing. *Checkpoint:* no bare fabric showing; rock clear of post bases by an inch or two.

Step 14 – Border and breaker support framing check. Already framed in Step 10; now verify the breaker blocking row is centered at X = 15'-6" from the finished corner and the picture-frame blocking is flush to the joist tops (plane/shim any block that sits proud – a proud block telegraphs through the border forever).

Step 15 – Deck-board layout. Measure the actual average board width from your delivered stack (10-board sample). Re-run the Section 7.1 math if it isn't 5 1/2 in. Snap the border lines on all four sides at exactly 5 1/2 in in from the finished edge; snap the breaker centerline at X = 15'-6"; snap a control line every 5 courses across the joists. *Mistake:* starting the field against a crooked border – string the border first.

Step 16 – Field decking. Start at the Y = 0 border line and work toward Y = 21 ft. Bark side / crown down, best face up; 3/32-in gaps (9/32 at the two borders, Sec. 11.4); ends cut at 14'-9 9/16" die into the border and breaker gaps; predrill all ends; 2 screws per bearing, 3/4-1 in from edges, screws snugged flush (not buried – wet PT craters). Check the running dimension every 5 courses against the control lines and correct the gap 1/32 in at a time. *Mistake:* letting course lines wander, then "fixing" with a wedge-shaped board – the 5-course check prevents it. *Safety:* PT sawdust – mask when gang-cutting; wash hands before eating.

Step 17 – Center breaker board. Drop the breaker into its gap over the blocking row (pieces 16'-0" + 4'-0 13/16", butt centered on a block), predrill, 2 screws per 16 in. Its edges get the same 3/32-in gap as everything else.

Step 18 – Picture-frame border. Fascia goes on FIRST (Step 19) if you want the border to conceal the fascia's top edge – recommended order is actually 19 then 18; they are listed per your sequence, and both orders work because the border overhangs nothing: install the four border runs to the snapped lines, miters tight and predrilled, 2 screws at every joist/blocking bearing (16 in o.c.). The border's outside edge must land exactly on the finished 31 x 21 line – string it, don't trust the frame. *Mistake:* screwing the miter tips without pilots (they split).

Step 19 – Fascia. Hang the 2x10 fascia with its top edge tight to the underside of the decking plane (+23 in / +22 1/2 in) so the border sits flush on top of it; SS trim screws 2 per 16 in; 1/8-in gaps at butts; corners closed to the 31/21 lines; seal every cut. (If you follow the recommended 19-before-18 order, tack the fascia 1/16 in high and let the border push it flush.)

Step 20 – Closeout. Hardware check: every ABU/BCS/H2.5 hole filled, anchors tight, lap nails in, no shiners at the rims. Re-treat any missed cut. Sweep gaps clear, clear gravel drips, haul offcuts. Walk the surface: no screw heads proud, no corner rock. Verify final numbers one last time: 31'-0" x 21'-0" outside decking; +24 in surface; diagonals 37'-5 5/16". Request the final inspection (816-228-0104, by 3 p.m. the day before) if a permit was issued, and keep the approval with your house records. # 13. Materials-Only Cost Estimates

Pricing checked: July 15, 2026. Read this honesty statement first:

- Home Depot, Lowe's, and Menards render prices only in a live browser session and block automated per-item price retrieval, and store-level stock for Blue Springs (Home Depot Blue Springs #3021 area, Lowe's Blue Springs, Menards Independence / Lee's Summit) could not be verified programmatically on the check date. **Every unit price below is therefore an ESTIMATE** built from current publicly visible listings, recent verified reference points (e.g., Simpson ABU66Z at about $26 retail and H2.5AZ under $1.25 at national suppliers on the check date), and typical Kansas City-metro PT lumber pricing. Estimates are believed accurate to roughly +/-15% per line. **Get a written quote from each store before budgeting.**
- No retailer's price was copied to another retailer; each column was estimated independently from that retailer's own typical positioning (Menards generally lowest on commodity PT lumber in the KC market, Lowe's typically highest by a few percent). Where a retailer likely does NOT stock an item, it is flagged below.
- Prices exclude tax, labor, permit fees, delivery, and tools, and include the full 10% waste/overage quantities from Section 10.
- Gravel: big boxes only sell bagged (about 360 x 0.5-cu-ft bags, roughly $1,600-1,900 – not sensible). The gravel line uses a **bulk local rock-yard allowance of $38/ton**, identical in every column and clearly not a big-box price.

Stock-status flags (as typically true in the KC metro – verify locally):

- Simpson BCS2-3/6Z: frequently **special order** at Home Depot and Lowe's; Menards often sells the USP/MiTek equivalent (acceptable if load-rated equal – verify rating). Order 2-3 weeks ahead.
- 2x8x10 GC and 5/4x6x8: stocked, but GC (ground-contact) versions of short lengths are sometimes above-ground only – **check the tag for UC4A/ground contact**; do not substitute above-ground treatment for beams, posts, or joists.
- 6x6x8 GC, 80-lb concrete, ABU66Z, H2.5AZ, deck screws, fabric: normally stocked for pickup at all three chains.
- Menards note: the frequent 11% rebate (store credit) is NOT reflected – when running, it effectively widens Menards' lead; also ask all three Pro Desks for a volume/bid quote on the whole list ("bundle" pricing often beats line-item shopping even before delivery, which you excluded).
- With the estimated prices, Menards is cheapest on every line, so the combined basket equals the Menards basket; with real quotes some lines (especially Simpson hardware at Home Depot) may flip – re-run the comparison with your quotes.

Product links (all verified live on July 15, 2026 as product/category pages; prices not displayed to automated access): Home Depot 5/4x6x16 GC SYP decking (homedepot.com/p/206966950), Home Depot PT lumber (homedepot.com/b/Lumber-Composites-Pressure-Treated-Lumber/N-5yc1vZc3sr), Home Depot ABU66Z (homedepot.com/p/100374896), Lowe's 80-lb concrete (lowes.com/pd/3006075), Menards treated lumber (menards.com/main/building-materials/lumber-boards/treated-wood-products/c-13123.htm), Menards AC2 2x10x16 GC (menards.com item 1111668). Search each retailer's site for the remaining line items – descriptions below are written to match their catalog wording.

Option A (5/4x6 decking) – Home Depot (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$15.87	$1,015.68
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$21.98	$461.58
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$13.48	$94.36
6x6x8 PT SYP ground contact (post blocks)	2 ea	$36.98	$73.96
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$7.25	$1,218.00
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$6.98	$118.66
Simpson ABU66Z standoff post base	15 ea	$23.68	$355.20
Simpson BCS2-3/6Z post cap	15 ea	$19.97	$299.55
Simpson H2.5AZ tie	55 ea	$1.08	$59.40
Simpson SD9112R500 connector screws (500)	2 box	$44.97	$89.94
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$51.98	$103.96
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$27.98	$27.98
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$61.98	$61.98
Copper naphthenate end-cut preservative, qt	2 ea	$20.98	$41.96
2x10x16 PT SYP ground contact (fascia)	7 ea	$28.98	$202.86
2x10x8 PT SYP ground contact (fascia)	2 ea	$14.98	$29.96
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$68.98	$206.94
6-in fabric staples (100 ct)	1 box	$12.98	$12.98
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
5/4x6x16 PT SYP deck board, ground contact	103 ea	$17.98	$1,851.94
5/4x6x8 PT SYP deck board	4 ea	$9.48	$37.92
#9 x 2-1/2 in exterior deck screws, 25-lb pail	1 pail	$109.00	$109.00
#9 x 2-1/2 in exterior deck screws, 5-lb box	1 box	$31.98	$31.98

Category subtotal	Amount
Structural framing	$1,645.58
Decking	$2,030.84
Concrete and footings	$1,336.66
Hardware and fasteners	$977.99
Fascia	$294.80
Landscape fabric and gravel	$573.32
TOTAL (Home Depot, estimated)	**$6,859.19**

Option A (5/4x6 decking) – Lowe's (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$16.35	$1,046.40
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$22.75	$477.75
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$13.85	$96.95
6x6x8 PT SYP ground contact (post blocks)	2 ea	$38.45	$76.90
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$7.15	$1,201.20
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$7.25	$123.25
Simpson ABU66Z standoff post base	15 ea	$24.98	$374.70
Simpson BCS2-3/6Z post cap	15 ea	$21.48	$322.20
Simpson H2.5AZ tie	55 ea	$1.15	$63.25
Simpson SD9112R500 connector screws (500)	2 box	$46.25	$92.50
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$53.50	$107.00
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$28.75	$28.75
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$63.50	$63.50
Copper naphthenate end-cut preservative, qt	2 ea	$21.75	$43.50
2x10x16 PT SYP ground contact (fascia)	7 ea	$29.85	$208.95
2x10x8 PT SYP ground contact (fascia)	2 ea	$15.45	$30.90
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$71.50	$214.50
6-in fabric staples (100 ct)	1 box	$13.45	$13.45

Item	Qty	Unit price (est.)	Extended
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
5/4x6x16 PT SYP deck board, ground contact	103 ea	$18.45	$1,900.35
5/4x6x8 PT SYP deck board	4 ea	$9.75	$39.00
#9 x 2-1/2 in exterior deck screws, 25-lb pail	1 pail	$112.00	$112.00
#9 x 2-1/2 in exterior deck screws, 5-lb box	1 box	$32.85	$32.85

Category subtotal	Amount
Structural framing	$1,698.00
Decking	$2,084.20
Concrete and footings	$1,324.45
Hardware and fasteners	$1,031.90
Fascia	$303.35
Landscape fabric and gravel	$581.35
TOTAL (Lowe's, estimated)	**$7,023.25**

Option A (5/4x6 decking) – Menards (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$13.99	$895.36
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$19.49	$409.29
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$11.99	$83.93
6x6x8 PT SYP ground contact (post blocks)	2 ea	$32.99	$65.98
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$6.79	$1,140.72
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$6.49	$110.33
Simpson ABU66Z standoff post base	15 ea	$21.99	$329.85
Simpson BCS2-3/6Z post cap	15 ea	$18.99	$284.85
Simpson H2.5AZ tie	55 ea	$0.89	$48.95
Simpson SD9112R500 connector screws (500)	2 box	$41.99	$83.98
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$46.99	$93.98

Item	Qty	Unit price (est.)	Extended
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$24.99	$24.99
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$54.99	$54.99
Copper naphthenate end-cut preservative, qt	2 ea	$18.99	$37.98
2x10x16 PT SYP ground contact (fascia)	7 ea	$25.99	$181.93
2x10x8 PT SYP ground contact (fascia)	2 ea	$12.99	$25.98
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$58.99	$176.97
6-in fabric staples (100 ct)	1 box	$10.99	$10.99
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
5/4x6x16 PT SYP deck board, ground contact	103 ea	$15.99	$1,646.97
5/4x6x8 PT SYP deck board	4 ea	$8.29	$33.16
#9 x 2-1/2 in exterior deck screws, 25-lb pail	1 pail	$96.98	$96.98
#9 x 2-1/2 in exterior deck screws, 5-lb box	1 box	$27.99	$27.99

Category subtotal	Amount
Structural framing	$1,454.56
Decking	$1,805.10
Concrete and footings	$1,251.05
Hardware and fasteners	$904.58
Fascia	$262.90
Landscape fabric and gravel	$541.36
TOTAL (Menards, estimated)	**$6,219.55**

Option A (5/4x6 decking) – Lowest-cost combined basket

Item	Qty	Supplier	Unit (est.)	Extended
2x8x12 PT SYP No.2 ground contact	64 ea	Menards	$13.99	$895.36
2x8x16 PT SYP No.2 ground contact	21 ea	Menards	$19.49	$409.29
2x8x10 PT SYP No.2 ground contact	7 ea	Menards	$11.99	$83.93
6x6x8 PT SYP ground contact	2 ea	Menards	$32.99	$65.98

Item	Qty	Supplier	Unit (est.)	Extended
80-lb high-strength concrete mix (4000 psi)	168 bag	Menards	$6.79	$1,140.72
Titen HD 5/8 x 6 in screw anchor	17 ea	Menards	$6.49	$110.33
Simpson ABU66Z standoff post base	15 ea	Menards	$21.99	$329.85
Simpson BCS2-3/6Z post cap	15 ea	Menards	$18.99	$284.85
Simpson H2.5AZ tie	55 ea	Menards	$0.89	$48.95
Simpson SD9112R500 connector screws (500)	2 box	Menards	$41.99	$83.98
10d x 3 in HDG ring-shank nails, 10 lb	2 box	Menards	$46.99	$93.98
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	Menards	$24.99	$24.99
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb	1 box	Menards	$54.99	$54.99
Copper naphthenate end-cut preservative, qt	2 ea	Menards	$18.99	$37.98
2x10x16 PT SYP ground contact	7 ea	Menards	$25.99	$181.93
2x10x8 PT SYP ground contact	2 ea	Menards	$12.99	$25.98
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	Menards	$58.99	$176.97
6-in fabric staples (100 ct)	1 box	Menards	$10.99	$10.99
3/4-in clean crushed limestone, BULK (local rock yard)	9.3 ton	Home Depot	$38.00	$353.40
5/4x6x16 PT SYP deck board, ground contact	103 ea	Menards	$15.99	$1,646.97
5/4x6x8 PT SYP deck board	4 ea	Menards	$8.29	$33.16
#9 x 2-1/2 in exterior deck screws, 25-lb pail	1 pail	Menards	$96.98	$96.98
#9 x 2-1/2 in exterior deck screws, 5-lb box	1 box	Menards	$27.99	$27.99

Category subtotal	Amount
Structural framing	$1,454.56
Decking	$1,805.10
Concrete and footings	$1,251.05
Hardware and fasteners	$904.58
Fascia	$262.90
Landscape fabric and gravel	$541.36
TOTAL (combined basket, estimated)	**$6,219.55**

Option B (2x6 decking) – Home Depot (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$15.87	$1,015.68
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$21.98	$461.58
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$13.48	$94.36
6x6x8 PT SYP ground contact (post blocks)	2 ea	$36.98	$73.96
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$7.25	$1,218.00
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$6.98	$118.66
Simpson ABU66Z standoff post base	15 ea	$23.68	$355.20
Simpson BCS2-3/6Z post cap	15 ea	$19.97	$299.55
Simpson H2.5AZ tie	55 ea	$1.08	$59.40
Simpson SD9112R500 connector screws (500)	2 box	$44.97	$89.94
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$51.98	$103.96
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$27.98	$27.98
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$61.98	$61.98
Copper naphthenate end-cut preservative, qt	2 ea	$20.98	$41.96
2x10x16 PT SYP ground contact (fascia)	7 ea	$28.98	$202.86
2x10x8 PT SYP ground contact (fascia)	2 ea	$14.98	$29.96
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$68.98	$206.94
6-in fabric staples (100 ct)	1 box	$12.98	$12.98

Item	Qty	Unit price (est.)	Extended
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
2x6x16 PT SYP No.2 ground contact	103 ea	$21.48	$2,212.44
2x6x8 PT SYP No.2 ground contact	4 ea	$10.98	$43.92
#9/#10 x 3 in exterior deck screws, 25-lb pail	1 pail	$115.00	$115.00
#9/#10 x 3 in exterior deck screws, 5-lb box	2 box	$34.98	$69.96

Category subtotal	Amount
Structural framing	$1,645.58
Decking	$2,441.32
Concrete and footings	$1,336.66
Hardware and fasteners	$977.99
Fascia	$294.80
Landscape fabric and gravel	$573.32
TOTAL (Home Depot, estimated)	**$7,269.67**

Option B (2x6 decking) – Lowe's (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$16.35	$1,046.40
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$22.75	$477.75
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$13.85	$96.95
6x6x8 PT SYP ground contact (post blocks)	2 ea	$38.45	$76.90
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$7.15	$1,201.20
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$7.25	$123.25
Simpson ABU66Z standoff post base	15 ea	$24.98	$374.70
Simpson BCS2-3/6Z post cap	15 ea	$21.48	$322.20
Simpson H2.5AZ tie	55 ea	$1.15	$63.25
Simpson SD9112R500 connector screws (500)	2 box	$46.25	$92.50
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$53.50	$107.00

Item	Qty	Unit price (est.)	Extended
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$28.75	$28.75
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$63.50	$63.50
Copper naphthenate end-cut preservative, qt	2 ea	$21.75	$43.50
2x10x16 PT SYP ground contact (fascia)	7 ea	$29.85	$208.95
2x10x8 PT SYP ground contact (fascia)	2 ea	$15.45	$30.90
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$71.50	$214.50
6-in fabric staples (100 ct)	1 box	$13.45	$13.45
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
2x6x16 PT SYP No.2 ground contact	103 ea	$21.98	$2,263.94
2x6x8 PT SYP No.2 ground contact	4 ea	$11.25	$45.00
#9/#10 x 3 in exterior deck screws, 25-lb pail	1 pail	$118.00	$118.00
#9/#10 x 3 in exterior deck screws, 5-lb box	2 box	$35.75	$71.50

Category subtotal	Amount
Structural framing	$1,698.00
Decking	$2,498.44
Concrete and footings	$1,324.45
Hardware and fasteners	$1,031.90
Fascia	$303.35
Landscape fabric and gravel	$581.35
TOTAL (Lowe's, estimated)	**$7,437.49**

Option B (2x6 decking) – Menards (estimated, pre-tax, materials only)

Item	Qty	Unit price (est.)	Extended
2x8x12 PT SYP No.2 ground contact (joists/blocking)	64 ea	$13.99	$895.36
2x8x16 PT SYP No.2 ground contact (beam plies/rims)	21 ea	$19.49	$409.29
2x8x10 PT SYP No.2 ground contact (beam middle plies)	7 ea	$11.99	$83.93
6x6x8 PT SYP ground contact (post blocks)	2 ea	$32.99	$65.98

Item	Qty	Unit price (est.)	Extended
80-lb high-strength concrete mix (4000 psi) (piers)	168 bag	$6.79	$1,140.72
Titen HD 5/8 x 6 in screw anchor (post-base anchors)	17 ea	$6.49	$110.33
Simpson ABU66Z standoff post base	15 ea	$21.99	$329.85
Simpson BCS2-3/6Z post cap	15 ea	$18.99	$284.85
Simpson H2.5AZ tie	55 ea	$0.89	$48.95
Simpson SD9112R500 connector screws (500)	2 box	$41.99	$83.98
10d x 3 in HDG ring-shank nails, 10 lb	2 box	$46.99	$93.98
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	$24.99	$24.99
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb (fascia)	1 box	$54.99	$54.99
Copper naphthenate end-cut preservative, qt	2 ea	$18.99	$37.98
2x10x16 PT SYP ground contact (fascia)	7 ea	$25.99	$181.93
2x10x8 PT SYP ground contact (fascia)	2 ea	$12.99	$25.98
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	$58.99	$176.97
6-in fabric staples (100 ct)	1 box	$10.99	$10.99
3/4-in clean crushed limestone, BULK (local rock yard) (9.3 tons)	9.3 ton	$38.00	$353.40
2x6x16 PT SYP No.2 ground contact	103 ea	$18.99	$1,955.97
2x6x8 PT SYP No.2 ground contact	4 ea	$9.49	$37.96
#9/#10 x 3 in exterior deck screws, 25-lb pail	1 pail	$101.99	$101.99
#9/#10 x 3 in exterior deck screws, 5-lb box	2 box	$29.99	$59.98

Category subtotal	Amount
Structural framing	$1,454.56
Decking	$2,155.90
Concrete and footings	$1,251.05
Hardware and fasteners	$904.58
Fascia	$262.90
Landscape fabric and gravel	$541.36
TOTAL (Menards, estimated)	**$6,570.35**

Option B (2x6 decking) – Lowest-cost combined basket

Item	Qty	Supplier	Unit (est.)	Extended
2x8x12 PT SYP No.2 ground contact	64 ea	Menards	$13.99	$895.36
2x8x16 PT SYP No.2 ground contact	21 ea	Menards	$19.49	$409.29
2x8x10 PT SYP No.2 ground contact	7 ea	Menards	$11.99	$83.93
6x6x8 PT SYP ground contact	2 ea	Menards	$32.99	$65.98
80-lb high-strength concrete mix (4000 psi)	168 bag	Menards	$6.79	$1,140.72
Titen HD 5/8 x 6 in screw anchor	17 ea	Menards	$6.49	$110.33
Simpson ABU66Z standoff post base	15 ea	Menards	$21.99	$329.85
Simpson BCS2-3/6Z post cap	15 ea	Menards	$18.99	$284.85
Simpson H2.5AZ tie	55 ea	Menards	$0.89	$48.95
Simpson SD9112R500 connector screws (500)	2 box	Menards	$41.99	$83.98
10d x 3 in HDG ring-shank nails, 10 lb	2 box	Menards	$46.99	$93.98
16d x 3-1/2 in HDG ring-shank nails, 5 lb	1 box	Menards	$24.99	$24.99
#8 x 2-1/2 in 305 SS trim-head screws, 5 lb	1 box	Menards	$54.99	$54.99
Copper naphthenate end-cut preservative, qt	2 ea	Menards	$18.99	$37.98
2x10x16 PT SYP ground contact	7 ea	Menards	$25.99	$181.93
2x10x8 PT SYP ground contact	2 ea	Menards	$12.99	$25.98
Woven landscape fabric 4 ft x 100 ft (heavy)	3 roll	Menards	$58.99	$176.97
6-in fabric staples (100 ct)	1 box	Menards	$10.99	$10.99
3/4-in clean crushed limestone, BULK (local rock yard)	9.3 ton	Home Depot	$38.00	$353.40

Item	Qty	Supplier	Unit (est.)	Extended
2x6x16 PT SYP No.2 ground contact	103 ea	Menards	$18.99	$1,955.97
2x6x8 PT SYP No.2 ground contact	4 ea	Menards	$9.49	$37.96
#9/#10 x 3 in exterior deck screws, 25-lb pail	1 pail	Menards	$101.99	$101.99
#9/#10 x 3 in exterior deck screws, 5-lb box	2 box	Menards	$29.99	$59.98

Category subtotal	Amount
Structural framing	$1,454.56
Decking	$2,155.90
Concrete and footings	$1,251.05
Hardware and fasteners	$904.58
Fascia	$262.90
Landscape fabric and gravel	$541.36
TOTAL (combined basket, estimated)	**$6,570.35**

Factor	16 in o.c. (as designed)	12 in o.c.
Joist lines / pieces	25 / 50	32 / 64
Extra lumber + screws	–	+14 pcs 2x8x12 + approx. 25% more deck screws: roughly $260-300
Code compliance (5/4 SYP)	Yes (16 in is the 5/4 limit)	Yes
Structure stiffness	Already approx. L/970 vs L/360 minimum (Sec. 4.2) – the short 8'-6" spans, not spacing, dominate	Marginal gain
Decking feel between joists	5/4: good; 2x6: excellent	5/4: excellent; 2x6: overkill
Labor	50 joists, 2,450 screws	64 joists, approx. 3,000 screws

Quantified decking-deflection comparison (relative sag between joists, deflection scales with spanˆ4 / thicknessˆ3): 5/4 on 12-in centers = 20,736/1.0 = 20,700 (index); 2x6 on 16-in centers = 65,536/3.375 = 19,400 (index). **2x6 decking on 16-in centers is slightly STIFFER underfoot than 5/4 on 12-in centers, while using 14 fewer joists.**

Recommendation: 16-in on-center. If you want a firmer walking surface, spend the upgrade money on 2x6 decking (Option B), not on 12-in joist spacing – it buys more stiffness for similar dollars with less labor. This follows your instruction to favor the practical, economical layout rather than either extreme.

15. Final Comparison and Recommended Configuration

15.1 Option A (5/4x6) vs Option B (2x6)

Factor	Option A: 5/4x6	Option B: 2x6
Material cost (Menards-basis estimate)	approx. $6,220 total project	approx. $6,570 total project (+ approx. $350)
Stiffness underfoot	Good at 16 in o.c.	approx. 3.4x stiffer board; noticeably more solid
Weight	approx. 1.9 psf decking	approx. 2.6 psf (framing unaffected)
Fasteners	#9 x 2 1/2 in	#9/#10 x 3 in (slightly costlier)
Ease of installation	Lighter boards, easier to spring straight	Heavier; stiffer boards resist straightening bowed stock but hide joist crowns better
Durability	Radius-edge 5/4 wears and cups sooner; more prone to end splits	Thicker section checks less, holds screws better, longer service life
Availability	Universally stocked	Universally stocked
Appearance	Classic radius-edge deck board	Beefier, flatter look

15.2 Recommended final configuration

2x6 decking (Option B) on the 16-in o.c. 2x8 frame as designed: 3 beams (3-ply 2x8), 15 posts/piers, 5/8 anchors + ABU66Z + BCS2-3/6Z + H2.5AZ hardware chain, picture frame + breaker, 2x10 fascia, 3 in clean rock over woven fabric.

Reasoning against your stated goals: strength (every member at 25-40% span margin); modest stiffness upgrade (approx. L/970 structure + the stiffest common decking, for about 5% over the cheapest build); simplicity (one joist size, one beam recipe, 15 identical footings, no ripped decking course); common materials (nothing longer than 16 ft, everything a stocked SKU except one connector to pre-order); low total cost (approx. $350 over Option A on a approx. $6,200-6,600 project); durability (GC lumber throughout, stainless face screws on fascia, end-cut treatment, taped beam option). If the budget is fixed and every dollar counts, Option A at 16 in o.c. is still a fully code-compliant, good deck – nothing in this plan changes except the decking SKU, screw length, and post-block height.

16. Scheduling and Inspection Recap (Blue Springs specifics)

Inspections requests by 3 p.m. the prior business day: 816-228-0104. Likely stages if permitted: (1) footing/open-hole before pour; (2) framing before decking; (3) final. Permit questions: 816-228-0118. Confirm stages when the permit (or exemption) is issued.

17. Required Cautions – Read Before Building

- **Underground utilities MUST be located before excavation.** Call 811 (Missouri One Call) and wait for marks. Fifteen 36-in-deep holes is real excavation.
- **Final footing dimensions depend on verified soil-bearing capacity and local requirements.** The 18-in/20-in diameters assume 1,500 psf clay; Blue Springs must confirm the bearing value, diameters, and the 36-in depth before you dig or pour.
- **The Blue Springs building department must approve the final design** (or confirm in writing that the project is exempt from permit). The permit-exemption language and the City's tip sheet conflict (Sec. 2.1); do not resolve that conflict yourself.
- **A licensed structural engineer should review anything outside prescriptive provisions** – including the freestanding lateral-load path (Sec. 4.8), the stub-post connection detail, and the beam-on-pier alternative (Sec. 4.6), if the City asks.
- **Verify lumber species, grade, treatment (UC4A ground contact), and actual dimensions at purchase.** Span tables assume SYP No. 2; some stores stock SPF or "No. 2 prime" variants. A different species or above-ground treatment voids these tables. Measure a sample: 2x8 = 1 1/2 x 7 1/4 in actual.
- **Retail pricing and inventory change constantly.** All prices herein are estimates dated July 15, 2026 (Sec. 13) and must be re-quoted.

- **Do not use this plan if field conditions differ materially from the assumptions:** sloping or disturbed ground, fill soils, water in footing holes, existing utilities or easements under the footprint, future hot tub/roof/pergola loads, or any change to the 24-in height or 31 x 21 footprint.
- This plan was prepared by an AI assistant applying published code tables and standard practice. It is a construction-planning document, not a sealed engineering design.

18. Consistency Audit (performed before delivery)

Checks performed across all sections; all pass:

1. Geometry: finished 31'-0" x 21'-0" = frame 30'-9" x 20'-9" + 1 1/2-in fascia both directions (369 + 3 = 372 in; 249 + 3 = 252 in). Diagonals: finished 37'-5 5/16"; frame 37'-1 3/16". Decking width: 45 x 5.5 + 42 x 3/32 + 2 x 9/32 = 252.0 in exact. Decking length: 5.5 + 3/32 + 177 9/16 + 3/32 + 5.5 + 3/32 + 177 9/16 + 3/32 + 5.5 = 372.0 in exact.
2. Elevations: 2.5 + 1 + 5 + 7.25 + 7.25 + 1 = 24.00 in (Option A); 2.5 + 1 + 4.5 + 7.25 + 7.25 + 1.5 = 24.00 in (Option B).
3. Spans: joist 8'-6" vs 11'-10" allowed; joist cantilever 21 in vs 25 1/2 allowed; beam 7'-0" vs approx. 8'-11" allowed; beam cantilever 16 in vs 21 in allowed.
4. Counts cross-checked between Sections 4, 5, 9, 10, 11, and 13: 25 joist lines = 50 pieces; 120 blocks (48 from offcuts + 72 cut = 8 boards); 15 posts/piers/bases/caps; 50 H2.5AZ used of 55 bought; 153 min / 168 bought concrete bags; 103 + 4 deck boards both options; screw counts 2,432 min / approx. 2,700 bought.
5. Beam ply buy list (15 x 16 ft + 6 x 10 ft) matches the splice diagram and the cut list; every splice lands on a post.
6. Cost tables sum correctly by construction (generated arithmetically); the combined basket legitimately equals Menards under estimated prices (noted in Sec. 13).

Assumptions that could materially change the design if wrong (beyond Sec. 2.3): soil bearing below 1,500 psf (larger footings; engineer); frost depth deeper than 36 in (longer piers, more concrete); City rejecting the stub-post detail (use the beam-on-pier alternative in Sec. 4.6); actual board width off 5 1/2 in (re-run Sec. 7.1 before framing the breaker blocking); GC lumber unavailable in a listed length (re-plan that member, do not substitute above-ground stock).

19. Sources

City of Blue Springs – Codes Administration Division (adopted codes, effective June 1, 2026; inspection rules): https://bluespringsgov.com/222/Codes-Administration-Division. Blue Springs Municipal Code Ch. 500 (frost depth 36 in, design criteria, permit exemptions): https://ecode360.com/28249653 and https://ecode360.com/28249484. Ordinance 5480 (Feb 18, 2026 code adoption): https://ecode360.com/BL3243/laws/LF2617498.pdf. Permit and Application Center: https://www.bluespringsgov.com/1603/Permit-Application-Center. IRC deck provisions (R507 joist/beam/footing tables; R301.5 loads; R312.1.1 guards; R401.4.1 soil): ICC, https://codes.iccsafe.org. Simpson Strong-Tie connector data: https://www.homedepot.com/p/100374896 (ABU66Z), https://shop.hancocklumber.com/product/simpson-bcs2-3-6z-post-base-cap-base-z-max (BCS2-3/6Z). Retail catalog references as listed in Section 13. Code and pricing information checked July 15, 2026.